Michael Zhan

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Education

- **Harvard University** **Sep 2023 – Dec 2024**
  M.S. Computational Science and Engineering (GPA: 3.96)
- **The University of Texas at Austin,** Engineering Honors Program **Sep 2019 – May 2023**
  B.S. Chemical Engineering, Elements of Computing Certificate (GPA: 3.99)

Professional Experience

Bloomberg | Software Engineer *New York, NY* **Jan 2025 – Present**

- Designed and implemented an attribute-based authorization engine in Go using AST-based policy representation and partial evaluation, compiling dynamic policies into optimized SQL filters in ~500µs to enforce access control across millions of entities.
- Engineered the Go backend of a content management platform across APIs, data modeling, storage, authorization, and content delivery; implemented access-controlled tree projections of cyclic, multi-parent collection graphs to retrieve hierarchical content.
- Led a large architectural refactor of a six-year-old Go CMS, improving separation of concerns and extracting shared code and interfaces, with a projected reduction of ~30K lines of redundant and boilerplate code.
- Built a data pipeline that transformed a heterogeneous graph of CMS content into a standardized, AI-compatible schema, deriving content-specific metadata such as hierarchical URLs to power natural-language retrieval on two Bloomberg websites serving tens of thousands of users monthly.

Amazon | Software Development Engineer Intern *Austin, TX* **May 2024 – Aug 2024**

- Extracted a critical onboarding microservice from a legacy monolithic architecture using Java and Spring dependency injection, reducing service onboarding time by 90% for over 10,000 internal AWS customers.
- Received full-time offer following internship.

Georgiou Research Group, UT Austin | Undergraduate Research Assistant *Austin, TX* **Jan 2020 – May 2023**

- Developed and implemented single-linkage clustering algorithms for >100,000 antibody sequence strings in Python and C++. Ran 50% faster than the previous internal tool, with identical results.
- Built k-means clustering workflow with Python scikit-learn and R to discover new gene segments in ferrets using outlier detection. Discovered 2 new genes with this method.

Genentech, Inc. | Software Engineer Intern *San Francisco, CA* **Jan 2022 – Dec 2022**

- Developed automation tooling for ML dataset generation and visualization on datasets with >500,000 samples using NumPy and pandas, reducing processing time by over 80%. In use across multiple Genentech sites.
- Engineered soft sensor machine learning models for cell culture parameter prediction using scikit-learn and AutoML libraries with an emphasis on feature optimization. Attained R^2 values of over 0.95.

Personal Projects

Stock Data Streaming System **Jul 2026**

- Built a real-time stock data pipeline in Go, simulating prices with geometric Brownian motion and streaming data to a frontend via Server-Sent Events (SSE).
- Implemented an O(1)-amortized sliding-window max algorithm to track rolling price ranges in real time and visualized streaming data using ChartGPU.

Mock Log System **Jun 2026**

- Built a distributed log-processing pipeline in C++ using Kafka, TimescaleDB, and PgBouncer, sustaining 20,000 logs/sec by decoupling log production and consumption and using async write batching via PostgreSQL streams.
- Implemented a custom HTTP server in C++ to handle concurrent frontend queries for time-series analytics.

Custom Language Interpreter **Nov 2025**

- Implemented lexing, Pratt parsing, and AST evaluation in Go for a custom language with primitives, functions, arrays, and hash maps.

Technical Skills

- **Languages:** Go, Python, C++, C, TypeScript, JavaScript, Java
- **Developer Tools:** Docker, Git, AWS, PostgreSQL, Kafka, CI/CD, Linux, GCP, Jira